

WASP-39b

R-1085

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_250330 · created 2026-07-30T16:06:25+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460764.8751 +/- 0.0086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.02
Transit depth (Rp/Rs)^2	1.64 +/- 0.5 [%]
Orbital Inclination (inc)	87.8 +/- 1.06
Airmass coefficient 1 (a1)	0.985 +/- 0.012
Airmass coefficient 2 (a2)	0.011 +/- 0.048
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	#2 - [233, 351]
Optimal Aperture	3.58
Optimal Annulus	8.1
Transit Duration (day)	0.114 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.128 ± 0.02 vs 0.1457 (deviation 0.78σ)
Duration fit vs published	0.114 d vs 0.12 d (deviation 0.41σ)
Mid-time O–C	-6.0 min
Depth SNR	5.7
Residual scatter	1.18 %

Light curve

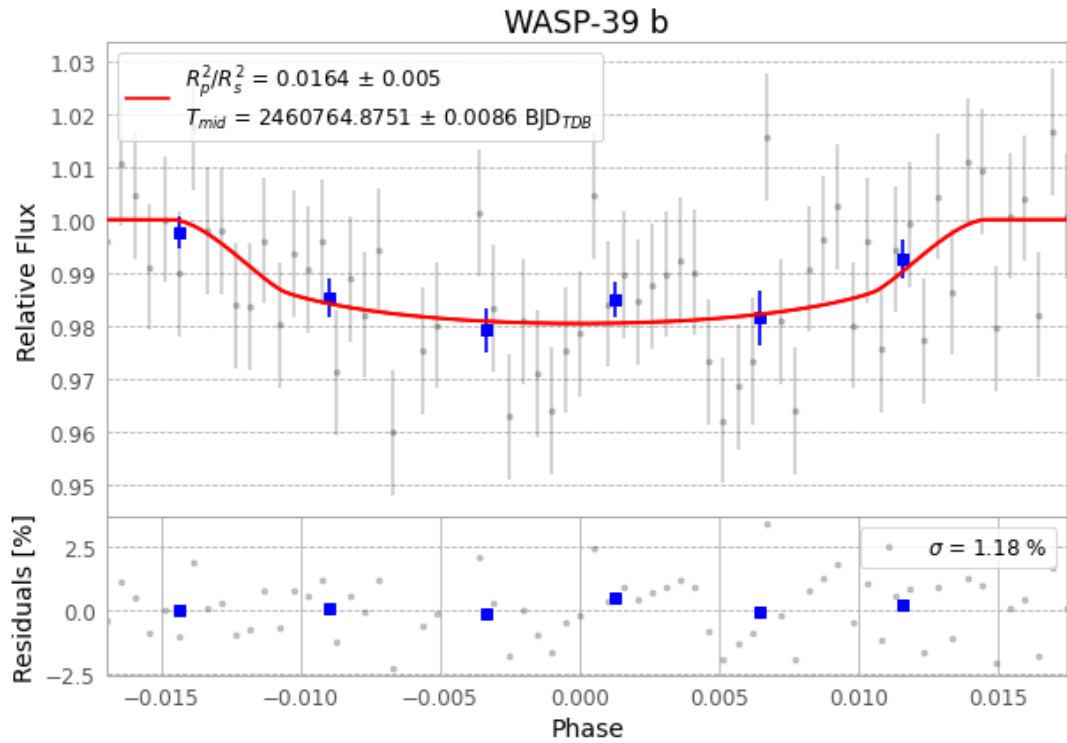


Figure 1 — FinalLightCurve_WASP-39 b_2025-03-30.png (SHA-256 d0620f9c80e0eb77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dc7fba5ebe961148...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), WASP-39250330071217.FITS → WASP-39250330103315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-250330.log (d774833f972d...), FinalLightCurve_WASP-39 b_2025-03-30.png (d0620f9c80e0...), FinalLightCurve_WASP-39 b_2025-03-30.csv (841e50469e10...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 16:06Z.